I. Use Case Description	
Use Case Name	Reasoning About Action Feasibility from Physical Properties
Use Case Identifier	UC-ActionFeasibility-01
Source	LACAI Lab, Department of Computer Science, Rensselaer Polytechnic Institute
Point of Contact	Hyejin Park, Department of Computer Science, Rensselaer Polytechnic Institute, parkh12@rpi.edu
Creation / Revision Date	September 11, 2026
Associated Documents	Nothing at this stage

II. Use Case Summary	
Goal	Develop an ontology that captures relationships among objects, their physical properties, and action requirements so that a system can reason about whether an object is suitable for a given action and explain why.
Requirements	The ontology should represent *objects, relevant object parts, selected manipulation actions, physical properties, and action requirements*, together with the relationships among them. For the selected actions, it should support reasoning about whether the properties of an object or object part satisfy the physical requirements of an action and provide a property-based explanation for the resulting feasibility judgment. It should also support retrieving candidate objects whose represented physical properties satisfy the requirements of a given action, rather than relying only on conventional object-action associations.
Scope	This use case focuses on affordance reasoning for an ordinary human actor interacting with everyday manipulable objects in a kitchen environment. The initial ontology will cover a small subset of elementary manipulation actions drawn from an ongoing research dataset, including actions such as *cut, open, pour, and squeeze*. It will represent only coarse physical characteristics directly relevant to these actions, such as *object-part geometry, rigidity/flexibility, and containment-related characteristics*. Chemical composition, nutritional properties, visual appearance, broader environments, non-human actors, and specialized or industrial equipment are outside the scope of this course project.
Priority	High — representing action feasibility from physical properties is central to the intended reasoning task and provides the foundation for later extensions of the ontology.
Stakeholders	Primary stakeholders are researchers in knowledge representation, robotics, and embodied AI who study object affordances and action feasibility. Secondary stakeholders include developers of intelligent systems that need to reason about object use from physical properties.
Description	This use case considers a user or researcher who wants to reason about whether an ordinary human actor could use an available object to support a particular manipulation action in a kitchen environment. Instead of relying only on an object's conventional name or typical function, the system uses an ontology that connects object parts and physical properties with the requirements of selected actions. The use case supports two primary types of interaction. First, a user can provide a specific object and action and ask whether the object is suitable for that action and why. Second, a user can provide an intended action and ask which available objects have physical characteristics that satisfy its requirements. In the latter case, the results may include objects that are not normally associated with the action but are physically suitable for the task. The ontology therefore supports both *feasibility explanation* and *property-based candidate object retrieval* within the limited scope of the course project.
Actors / Interfaces	**Primary Actor – User/Researcher:** Submits queries about whether a given object can support a particular action and receives the resulting judgment and explanation. **System – Ontology-Based Reasoning System:** Matches the physical properties of the object against the requirements of the action and infers whether the action is feasible or infeasible. **Secondary Actor / Interface – Knowledge Base/Data Sources:** Supplies structured information about objects, their parts and physical properties, and action requirements.

Pre-conditions	The ontology contains representations of the selected kitchen manipulation actions, relevant object or object-part properties, and the physical requirements associated with those actions. The specified or candidate objects have some relevant physical-property information available, although this information may be incomplete
Post-conditions	The system returns a feasibility judgment or a set of candidate objects, together with an explanation based on the relevant physical properties and action requirements. If the available knowledge is insufficient to support a judgment, the system indicates that additional property information is needed rather than forcing a feasible or infeasible conclusion.
Triggers	The use case is triggered when a user submits either (1) a specific object-action query asking whether the object can support the action, or (2) an action-oriented query asking which available objects satisfy the physical requirements of the action.
Performance Requirements	
Assumptions	
Open Issues	

III. Usage Scenarios

Scenario 1: Feasibility Verification and Explanation

A user or researcher has a specific object and wants to know whether an ordinary human could use it to perform a selected manipulation action in a kitchen environment. For example, the user may ask whether a rigid plastic card can be used to cut a soft food item. The system identifies the relevant part of the object, such as the card edge, and retrieves its represented physical properties, including relevant edge geometry and rigidity. It then compares these properties with the physical requirements associated with the *Cutting* action. If the relevant requirements are satisfied, the system returns a feasible judgment together with a property-based explanation. If required information is missing, the system indicates that there is insufficient evidence rather than forcing a yes/no conclusion.

Scenario 2 — Goal-Driven Alternative Object Discovery Under Constraints

An ordinary user in a kitchen needs to perform an action in an unexpected or time-critical situation where the conventional tool is unavailable or unusable. Instead of searching by object category, the user asks which available objects could physically support the required action. For example, if a cutting tool is unavailable, the system can retrieve the physical requirements of *Cutting* and compare them with the properties of nearby objects. A rigid plastic card may be returned as a candidate for cutting a soft food item if its edge is sufficiently thin and rigid, even though cutting is not its conventional function. The system therefore supports goal-driven discovery of alternative objects based on physical feasibility rather than typical object use.

IV. Basic Flow of Events

Basic Flow Narrative:
The user begins by specifying an intended manipulation action and, depending on the usage scenario, either a particular object to evaluate or a set of available objects to search. The system identifies the physical requirements associated with the action and retrieves the relevant physical properties of the specified object or candidate objects from the knowledge base. It then compares the represented object properties with the action requirements. Based on this comparison, the system either evaluates the feasibility of the specified object-action pair or retrieves candidate objects that satisfy the relevant requirements. Finally, the system returns the result together with a property-based explanation.

Basic / Normal Flow of Events			
Step	**Actor (Person)**	**Actor (System)**	**Description**
1	User		Specifies an intended manipulation action
2	User		Provides either a specific object to evaluate or a set of available objects to search
3		Ontology Based Reasoning System	Retrieves the physical requirements associated with the selected action
4		Knowledge Base / Reasoning System	Retrieves relevant parts and physical properties of the specified or available objects.
5		Ontology-Based Reasoning System	Compares the represented object properties with the physical requirements of the action
6		Ontology-Based Reasoning System	Infers either the feasibility of the specified object-action pair or candidate objects satisfying the action requirements
7		System	Returns the result and a property-based explanation to the user

V. Alternate Flow of Events

Alternate Flow Narrative — Insufficient Property Information

If the knowledge base does not contain sufficient physical-property information for an object or object part needed to evaluate an action requirement, the system cannot make a supported feasibility judgment. Instead of treating missing information as evidence of infeasibility, the system reports which required information is unavailable and returns an insufficient-information result. The user may then provide additional information, select another object, or end the query.

Alternate Flow of Events			
Step	**Actor (Person)**	**Actor (System)**	**Description**
A1		Reasoning System	Detects that one or more properties required for evaluating the action are missing from the knowledge base
A2		Reasoning System	Does not infer feasibility or infeasibility from the missing information
A3		System	Reports the missing property information and returns an insufficient-information result
A4	User		Provides additional information, chooses another candidate object, or ends the query.
A5		System	If sufficient information becomes available, resumes the normal flow at the property-comparison step

VI. Use Case and Activity Diagram(s)

Provide the primary use case diagram, including actors, and a high-level activity diagram to show the flow of primary events that include/surround the use case. Subordinate diagrams that map the flow for each usage scenario should be included as appropriate

VII. Competency Questions

Scenario 1 — Feasibility Verification and Explanation
CQ1.1 — *Can a rigid plastic card be used to cut a soft food item, and why?*

- Example Answer: Yes. A rigid plastic card may be suitable for cutting a soft food item if its edge is sufficiently thin and rigid to separate the material.
- Ontology Approach: The ontology represents classes such as *Object, ObjectPart, Action, PhysicalProperty, and ActionRequirement.* The card is linked to its edge through hasPart, and the edge is linked to relevant properties through *hasProperty*. The *Cutting* action is linked to its required physical properties through

requiresProperty. The system compares the represented properties of the card edge with the requirements of *Cutting* and returns a feasibility judgment with the properties that support the result.

CQ1.2 — *Can a sponge be used to cut a firm vegetable, and why or why not?*

- Example Answer: No, based on the represented properties. A typical kitchen sponge lacks the rigidity and suitable edge geometry required to separate a firm vegetable
- Ontology Approach: The system retrieves the physical requirements represented for *Cutting* and compares them with the relevant properties of the sponge. If a required property is explicitly incompatible with the represented properties of the object, the system can return an infeasible judgment and identify the unmet requirement. If the required property information is missing rather than incompatible, the result should instead be reported as insufficient information.

Scenario 2 — Goal-Driven Alternative Object Discovery Under Constraints

CQ2.1 — *If a conventional cutting tool is unavailable, which available objects could satisfy the physical requirements for cutting a soft food item?*

- Example Answer: Among a rigid plastic card, a rolling pin, and a sponge, the rigid plastic card may be returned as a candidate because its edge can provide the thin geometry and rigidity relevant to cutting a soft material.
- Ontology Approach: The system begins with the *Cutting* action and retrieves its associated *ActionRequirement* instances and required *PhysicalProperty* values. It then searches the available *Object* and *ObjectPart* instances for represented properties that satisfy those requirements. Candidate objects are therefore retrieved from physical compatibility rather than from a predefined list of conventional cutting tools.

CQ2.2 — *If a pitcher is unavailable, which available kitchen objects could satisfy the physical requirements for pouring water into a narrow container?*

- Example Answer: A small cup may be returned as a suitable candidate if it can contain the liquid, can be grasped and tilted, and has an opening that allows controlled transfer into the target container
- Ontology Approach: The ontology retrieves the requirements represented for *Pouring*, such as the ability to contain liquid and physical characteristics that support controlled transfer. It then identifies available objects whose represented parts and properties satisfy those requirements. The answer can therefore include objects that are physically suitable for pouring even when *Pouring* is not their primary conventional function.

VIII. Resources

Data Availability: EPIC-KITCHENS-100 is publicly available and provides the initial kitchen-domain action vocabulary. The physical-property, feasibility, and counterfactual relations required for the ontology are not directly available in EPIC-KITCHENS and are being generated as part of ongoing research. For this course project, a small curated subset covering the selected actions will be available and used to construct and test the ontology. The broader research dataset is outside the scope of this use case.

Knowledge Bases, Repositories, or other Data Sources

Data	Type	Characteristics	Description	Owner	Source	Access Policies & Usage
EPIC-KITCHENS-100	Remote dataset / repository	Egocentric kitchen videos with annotated manipulation actions and object interactions	Provides the kitchen-domain action vocabulary and examples of human-object interactions used to define the initial action scope of the course ontology. Only a small subset of actions is used in this project.	EPIC-KITCHENS / University of Bristol and collaborators	EPIC-KITCHENS dataset website / annotation repository Link (full dataset): https://epic-kitchens.github.io/ Link (annotation dataset): https://github.com/epic-kitchens/epic-kitchens-100-annotations	Publicly available for research use; Required for establishing the initial kitchen-domain action vocabulary.
Derived Feasibility Scenario Annotations	Local / experimentally generated dataset	Curated examples of feasible and infeasible object-action combinations for the selected kitchen actions, including relevant object parts and physical properties	Provides the instance-level evidence used to populate the course ontology with examples of objects, object parts, relevant physical properties, and feasible or infeasible outcomes. A small subset corresponding to the selected actions will be used for this course project.	Hyejin Park / LACAI Lab	Generated through ongoing affordance-reasoning experiments	Internal and under development. A curated subset will be available for use during the course; Required for populating and testing the ontology
Derived Counterfactual Contrast Annotations	Local / experimentally generated dataset	aired examples in which changes to an object property or relevant part are associated with changes in action feasibility	Provides supporting evidence for identifying which physical properties or object-part characteristics are relevant to an action requirement. For the course	Hyejin Park / LACAI Lab	Generated through ongoing affordance-reasoning experiments	Internal and under development; a limited subset will be available for the course project.

			project, only contrasts involving the selected actions and properties will be considered			Supporting resource for refining action requirements and property-based explanations

External Ontologies, Vocabularies, or other Model Services

Resource	Language	Description	Owner	Source	Describes/Uses	Access Policies & Usage
Word Net	English	Provides word senses, synonym sets, and lexical-semantic relations for English nouns and verbs. It can support normalization and sense distinction for action and object terms used in the ontology.	Princeton University	Princeton WordNet Link: https://wordnet.princeton.edu/	Action and object terminology derived from EPIC-KITCHENS and the experimental annotations	Freely available; Supporting / Optional

Other Resources, Service, or Triggers *(e.g., event notification services, application services, etc.)*

Resource	Type	Description	Owner	Source	Access Policies & Usage
(sensor or external service name)		*Include a description of the resource as well as availability, if applicable*	*Primary owner of the service*	***Application or service URL**; if subscription based, include subscription and any subscription owner*	

IX. References and Bibliography

List all reference documents – policy documents, regulations, standards, de-facto standards, glossaries, dictionaries and thesauri, taxonomies, and any other reference materials considered relevant to the use case

X. Notes

There is always some piece of information that is required that has no other place to go. This is the place for that information.

* LLM Use Statement:

I used ChatGPT (GPT-5.6 Sol) to help revise the English wording and organization of this document. All substantive content and ontology design decisions are my own.